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RESEARCH

Bow and Arrow Technology in North America

Raven Garvey[1] · Ian Beggen[1] · Rocky Brockway[2] · Christopher Morgan[2] · Shannon Tushingham[3] · Robert Bettinger[4]

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Abstract
The adoption and spread of bow and arrow technology in North America reflect a complex interplay of ecological and social factors: While environmental variables such as wood availability and prey diversity/behavior were surely important, demographic and cultural variables—including population size, density, and connectivity; cultural transmission processes; and social dynamics—were equally or more influential. Parsing the relative effects of these factors and understanding interactions among them requires a clear view of the timing and nature of bow use across North America's diverse geography. This paper makes two primary contributions to our understanding of the bow's adoption in North America. Firstly, we present evidence for the bow's earliest appearance, use in conjunction with other projectile technologies, and effects on economic and other systems in the North American Arctic, Pacific Northwest and Plateau, California and the Great Basin, Southwest, and Southeast. Secondly, we present a novel model of technological investment (uptake) that considers the effects of transmission agents' social roles: Whether agents are craft specialists or do-it-yourself tool producers–users affects rates of adoption, a finding with global implications demonstrated here through regional case studies. We conclude that adoption depends not just on the bow's inherent utility but on how tools are produced, shared, used, and valued in different economic systems.

Keywords Craft specialization · Cultural transmission · Diffusion of innovations · Technological intensification

✉ Raven Garvey
garveyr@umich.edu

[1] University of Michigan, Ann Arbor, USA

[2] University of Nevada, Reno, USA

[3] California Academy of Sciences, San Francisco, USA

[4] University of California, Davis, USA

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Introduction

Adoption of the bow and arrow is among the most significant technological transitions in ancient North America. As an innovation, it spread quickly across most ecoregions, and its adoption had far-reaching implications for subsistence and mobility strategies, social organization, and intergroup dynamics (Blitz, 1988). Distinguishing consequences of bow adoption from causes is challenging but essential to our understanding of both local histories and larger cultural evolutionary processes.

Studying the adoption of the bow at local scales is necessary because technological change is ultimately a product of decisions made by individuals and small groups within specific ecological and social contexts (Richerson and Boyd, 2005). Meanwhile, assessing the phenomenon at regional and global scales provides insights into macroevolutionary processes, *e.g.*, how the bow contributed to the emergence, persistence, and transformation of complex cultural systems over long timescales (Paige and Perreault, 2024). To understand the spread and influence of the bow, then, we must account for the ecological and social contexts of adoption on the one hand and attempt to address both microevolutionary (*e.g.*, individual- and local-level) and macroevolutionary (higher-level) phenomena on the other. At both scales, the study of bow adoption requires attention not only to the practical advantages the technology affords people on the ground but also to the structure of knowledge transmission itself.

Here, we take stock of current evidence for the bow's spread in North America. We examine the timing, speed, and context of its adoption across broad geographic regions, assessing ecological and social factors that shaped its diffusion. We also offer a novel model of technological investment that considers the effects of differences among agents of cultural transmission on the timing and speed of uptake. Adoption, we contend, reflects not just what the bow could do, but how different societies organized labor, distributed technological knowledge, and attached meaning to the innovation.

Economic and Social Implications of Bow and Arrow Adoption

Economic implications of bow adoption were both immediate and long-term (*e.g.*, Blitz, 1988). Hunting efficiency increased immediately; relative to the atlatl and dart system, bows provide greater accuracy, higher rates of fire (reload capacity), and improved portability and ease of use in multiple terrain types, allowing hunters to target a wider range of prey species with increased success (Bettinger, 2015; Hughes, 1998; Reed, 2010; Tomka, 2013; Whittaker, 2010). This efficiency had downstream effects on broader patterns of subsistence and mobility, *e.g.*, higher hunting success rates may have indirectly freed time for other activities, such as plant resource collection (Bettinger, 2013, 2015).

The bow also affected social dynamics, particularly interpersonal and intergroup conflict where its longer effective range and rapid reload capacity gave users a distinct tactical advantage (Blitz, 1988). Similarly, the innovation would have facilitated

defense of specific resource patches, potentially leading to increased territoriality (Blitz, 1988; Hughes, 1998; Shott, 1997). Archaeological evidence including settlement fortification and increased skeletal trauma coincident with the introduction of the bow supports hypotheses centered on the bow's contribution to bellicosity, as we describe in later sections.

Group composition, leadership structures, and norm enforcement may also have been affected by the adoption of the bow. On the one hand, more efficient food procurement may have contributed to population growth, which in turn encouraged novel social structures to generate and manage sufficient resources, build and maintain infrastructure, and organize defense against outsiders (Bettinger, 2015; Smith, 2021). On the other hand, the bow may have become a means of enforcing social norms and reducing freeloading in societies with increasing divisions of labor (Bingham & Souza, 2009).

Anthropological Implications of Bow and Arrow Adoption

Granting the bow's profound impact on both the economic and the social, it remains an open question whether functional superiority (in subsistence or social arenas, or both) is sufficient to explain the uptake of the technology across varied North American (and other, global) ecologies. We don't think so; history abounds with accounts of superior technologies having failed during uptake for reasons related to cultural transmission or path dependence (Richerson & Boyd, 2005).

We argue, further, that the stakes of understanding bow adoption are high: understanding the adoption process affects how we theorize cultural change, human decision-making, and relationships between tools and institutions. At the microevolutionary level, whether to adopt the bow was an individual decision shaped by personal or household economics, material constraints, learning biases, and social pressures. These local dynamics help explain the prolonged coexistence of bows and atlatls and the uneven pace of adoption across regions.

At the macroevolutionary scale, the bow's spread reshaped settlement and subsistence systems, terms of conflict and cooperation, and political organization, for example. Probing these dynamics, we might better understand how cumulative, small-scale decisions contribute to large-scale structural change. Without empirically grounded accounts of micro-level processes, macroevolutionary models risk overgeneralization; without macroevolutionary framing, micro-level variability lacks broader explanatory power.

Methodological Challenges in Identifying Bow and Arrow Adoption

At all analytical scales, efforts to reconstruct the timing, speed, and nature of bow adoption face methodological challenges. Pinpointing arrival is notoriously difficult because preservation and sampling biases limit our ability to precisely date the appearance of any innovation. Additionally, since only the stone portion of projectile technologies typically survives in archaeological records, we are often left

to distinguish between dart and arrow points based on morphometrics and weight (Connolly and Baxter, 2025).

There are many excellent archaeological and experimental studies that provide guidelines for distinguishing arrow points from dart points (*e.g.*, Fenenga, 1953; Hildebrandt & King, 2012; Hughes, 1998; Shott, 1997; Thomas, 1978). While weight and various width attributes have emerged repeatedly as reliable means of differentiating point types, technological typologies are generally region-specific. We do not attempt to standardize discriminant criteria across regions considered in this meta-analysis, instead deferring to regional expertise.

Still, caveats abound even among regional experts, and some projectile points have surely been misclassified (Connolly and Baxter, 2025; Erlandson *et al.*, 2014; Hughes, 1998). Indeed, this may be true of many datasets, including ones on which this paper draws, and we acknowledge that chronologies may change as new data emerge. Here, we offer updated regional syntheses based on current best evidence for the bow's initial appearance and spread, with the goal of clarifying previously described patterns and identifying new ones.

Regional Chronologies and Contexts of Adoption

To better understand the factors influencing bow and arrow adoption, we examine its spread across broad geographic regions, each characterized by distinct ecological and social conditions (Figs. 1 and 2). We acknowledge that our divisions may mask important patterns within and between regions and that a continental-scale (non-regional) assessment of dated bow sites could be revelatory. Such a study is beyond the scope of the current paper, which we believe nonetheless provides compelling evidence for relevant causes and consequences of bow adoption across the continent.

North American Arctic

The earliest evidence of bows in North America is found in the Arctic. Arrival of the technology may date to the early Holocene based on evidence from southwest Alaska, where bone points slotted for microblades have been directly dated to ca. 12,000 BP (Ackerman, 1964, 1992, 2011; Dixon, 2011). However, apart from a handful of sites dating closer to 9000 BP, evidence of bow use in the Arctic is scarce until 4500 BP (Larsen, 1968, in Maschner & Mason, 2013; Stanford, 1976).

Even those skeptical of the early Holocene evidence acknowledge that bow use in the Arctic significantly predates its widespread adoption in more southerly regions. Well-attested bow use is evident by ca. 4500 BP, when it was (re)introduced via the Bering Strait region as part of the Arctic Small Tool tradition (Anderson, 1988; Maschner & Mason, 2013). For example, well-preserved sites in western Greenland (*e.g.*, Qeqertasussuk and Qajaa) produced wooden arrow shafts, foreshafts, and bifacial stone points indicative of bow technology (Grønnow, 2012, 2017). Elsewhere, evidence is largely in the form of microliths assumed to be part of composite points

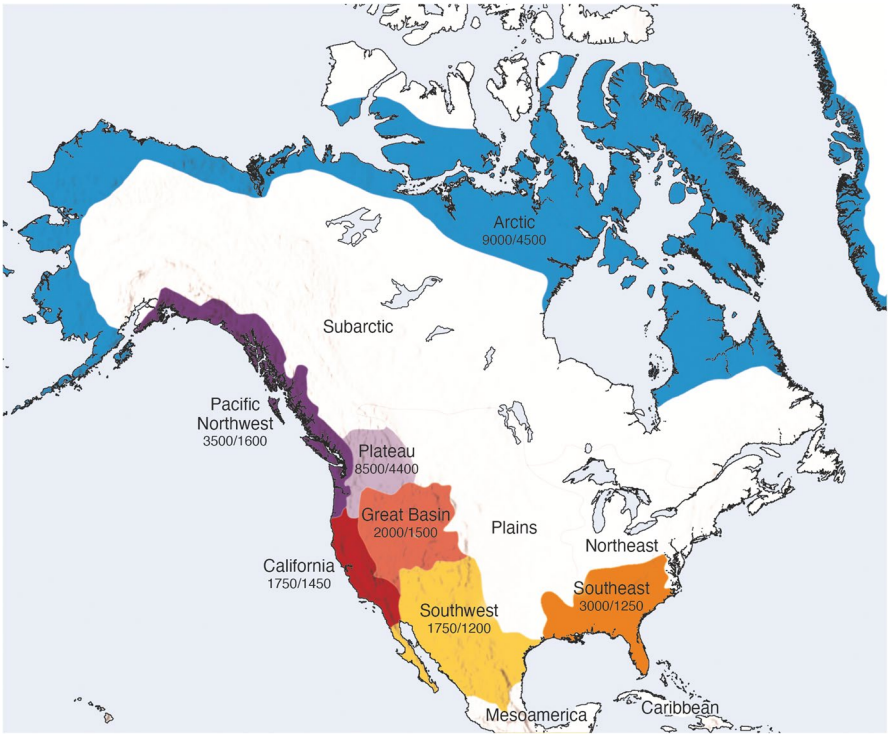

Fig. 1 Regions within North America described in this paper. Dates below each regional label (*e.g.*, "Plateau") indicate, in years cal BP, the earliest suggested date of bow adoption (before the backslash) and accepted date of adoption or date by which use is widespread (after backslash). See text for detailed discussions of dates. Adoption dates are omitted for the Subarctic, Plains, and Northeast because these regions are not discussed in the text

that may have developed in response to a general scarcity of knappable stone across the region (Andrefsky, 1994; Yi *et al.*, 2013).

Maschner and Mason (2013) report that, in Alaska, bow technology appears to drop out of the record ca. 3500 BP, coincident with a decline in caribou populations and a (likely related) reduction in archaeological sites across the interior. When bow technology reappears ca. 2500 BP, it may have been in the context of warfare (Workman, 1969), evidence for which is clear by 1300 BP, when innovations such as the recurved bow, armor, and fortified sites reflect the arrival of the Asian War Complex (Maschner & Mason, 2013).

Increasingly, ice patch archaeology is improving local cultural chronologies, including fluctuations in bow use. For example, melting snowfields in interior Yukon have produced well-preserved wooden arrow shafts, antler foreshafts, fletching, and sinew bindings (Hare *et al.*, 2012). Direct dates on these organic components suggest abrupt replacement of the atlatl by the bow ca. 1200 BP.

Ecologically, the bow and arrow were well suited to terrestrial Arctic hunting. Among the Thule, for example (A.D. 1200–A.D. 1900), muskox and, particularly,

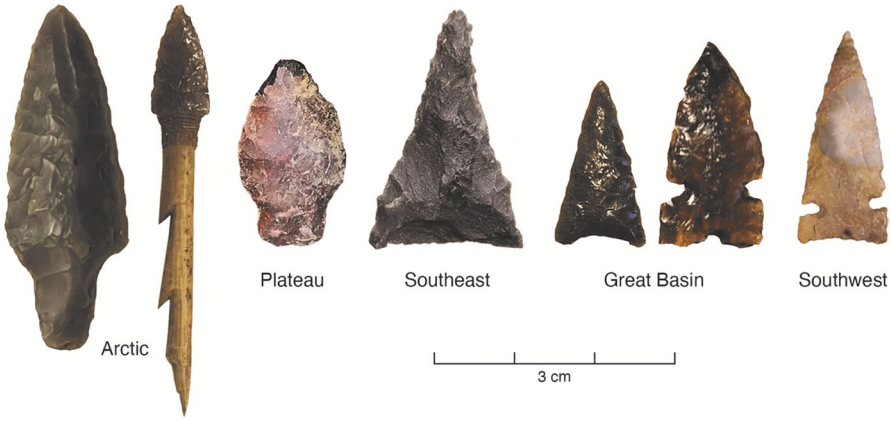

Fig. 2 Composite photographic representation of point styles observed in regions discussed in the text. Arctic (left): Alaska, Liebes Collection, California Academy of Sciences, photographed by Surya Kalaimani with permission from the California Academy of Sciences; Arctic (right): Alaska, California Academy of Sciences, photographed by Surya Kalaimani with permission from the California Academy of Sciences; Plateau: Chiliwist Phase (2200–3300 BP) point from site 45OK436, photographed by Adam Rorabaugh; Southeast: Pee Dee Triangular point from the Doerschuk site, North Carolina, photographed by I. Randolph Daniel; Great Basin (both): Desert Series arrow points, replicas knapped by Dan Steuber and photographed by Christopher Morgan; Southwest: Washita point from the Henderson site, New Mexico, photographed by Raven Garvey

caribou were frequently targeted with bows (Pfeifer, 2021). In Arctic coastal contexts, however, atlatl and harpoons remained important for hunting large marine mammals and were never replaced by the bow (Maschner & Mason, 2013), though bows were effective for hunting the small game and birds that supplemented marine-based diets.

As with knappable stone, many Arctic environments lack the high-quality, elastic wood necessary for effective bow production. In response, some Indigenous Arctic groups (*e.g.*, Greenlandic Thule) produced cable-backed bows, which used staves made from driftwood, antler, or whalebone reinforced with sinew cords stretched tightly across the back (Pfeifer, 2021). The sinew backing stored energy and helped prevent brittle staves from breaking under tension. The presence of similar, cable-backed bows among the White Mountain, Warm Springs, and Huachuca Apache in the Southwest, but nowhere else south of the Arctic/Sub-Arctic, almost certainly reflects the retention of the technology by Apacheans as they moved from the Sub-Arctic into the Southwest (Gifford, 1940: 29 l; Driver & Massey, 1957:351).

Social implications of Arctic bow use are also noteworthy. For example, oral traditions and historical accounts from Yup'ik groups in western Alaska describe the use of bows in interpersonal and intergroup conflict. The so-called "Bow and Arrow Wars" of the seventeenth century, recorded in Yup'ik oral histories, suggest that the bow played an important role in territorial defense and raiding during the Little Ice Age, which may have exacerbated existing tensions and resource stress (Fienup-Riordan, 1995).

Pacific Northwest and Plateau

There is widespread agreement that bow and arrow technology was introduced in the Interior Plateau earlier than elsewhere in temperate North America (Blitz, 1988; Chatters, 2004; Rousseau, 2004). This arrival, around 2500–2300 cal BP, is indicated by the appearance of small, thin, narrow-necked triangular points. A much earlier timing is provocatively suggested by Ames and colleagues (2010), who argue that some early Cascade points were used to tip arrows as early as 8500 cal BP in the Columbia Plateau. While initially rare, they suggest the bow was in widespread use by 4400 BP (5000 cal BP). Hatwai III points, dating to between 4400 and 2800 BP, are considered clear examples of arrow points (Ames *et al.*, 2010; but see Hildebrandt and King, 2012). By 1500–1300 BP, smaller arrow points are widespread, possibly associated with technological advances such as the sinew-backed bow (Bettinger *et al.*, 2006) or recurved bow (Mashner and Mason, 2013).

Adoption was slower on the Pacific Northwest Coast, but its use was widespread about 1600 BP based on the presence of small notched and notch triangular points made of chipped stone or ground slate, which are markers of the transition to the Late Pacific Period (Angelbeck & Cameron, 2014; Blitz, 1988; Matson & Coupland, 1995; Rorabaugh & Fulkerson, 2015). However, Erlandson and colleagues (2014) caution that the presence of small points is not unequivocal evidence of bow use, particularly in coastal regions and areas where aquatic foraging was prevalent over terrestrial mammal hunting. These small points may instead have been used to tip fishing implements such as harpoons or simple spears. Alternatively, as Connolly and Baxter (2025) argue for western Oregon's Willamette Valley, the co-occurrence of larger "dart-sized" and smaller "arrow-sized" points could reflect their use for different targets rather than in different delivery systems. Still, Rorabaugh and Fulkerson (2015), while cognizant of the limitations of inference from size alone, cautiously propose that the bow may have initially been in use as early as 3500 BP in the Salish Sea region, and that it became prevalent by 2500 BP (Rorabaugh & Fulkerson, 2015).

The bow was once widely assumed to have completely replaced the atlatl by 1000 BP (Ames *et al.*, 2010). However, mounting evidence suggests a more complex adoption process, with both weapon systems used simultaneously, to varying degrees, throughout the Northwest Coast and Plateau (see also Connelly and Baxter, 2025). In many places, bows and atlatls were complementary weapon systems, but arrows became dominant by 1000 BP.

Ames *et al.* (2010) propose a substantially extended period of co-use throughout the Holocene in the Columbia Plateau, characterized by shifting proportions of darts and arrows. Bows were rare in the early Holocene but became widespread by the Middle Holocene. After 3000 BP, arrows and darts became more differentiated, making the use of each system easier to identify archaeologically. After 1000 BP, atlatls were rare, but still in use.

Evidence for the persistent use of the atlatl is largely based on the presence of darts. However, a few exceptional finds of preserved wood and bone atlatls have been discovered in the region (Carlson & Magne, 2008). For instance, on the Pacific Northwest Coast, an atlatl dating to 1700 ± 100 BP was discovered at a site on the

Skagit River (Fladmark *et al.*, 1987), and numerous whalebone atlatls were recovered at the Par-tee site (35CLT20) on the Oregon Coast (Losey & Hull, 2019). Another example is an atlatl discovered at a site on Quiltanaton Lake in Central Fraser Plateau 1950 ± 100 BP (Carlson & Magne, 2008).

The bow and arrow introduced broad-reaching economic and social impacts in the region, reshaping hunting practices and community dynamics. Around 2500 BP, shifts in regional faunal assemblages indicate an intensification of deer and wapiti hunting, which became more productive and efficient (Angelbeck & Cameron, 2014; Butler & Campbell, 2004; Rousseau, 2004). In the Salish Sea region, the technology offered individual hunters a distinct advantage, allowing them to hunt more effectively, either independently or in smaller groups than are required for atlatl hunting. Increased autonomy enabled hunters to resist top-down elite control over collective hunting teams and other facets of community life (Angelbeck & Cameron, 2014; Rorabaugh, 2019). Furthermore, it may have served as a response to centralized political systems and escalating warfare during this period (Rorabaugh, 2019).

The bow and arrow also emerged as a more potent weapon due to its greater stealth, ability to facilitate ambushes, and improved accuracy, which contributed to heightened warfare and intergroup conflict. In the Pacific Northwest Coast, the Late Pacific Period is marked by an increase in warfare, evidenced by dated defensive sites fortified with palisades (Angelbeck & Cameron, 2014; Rorabaugh, 2019). On the Plateau, Chatters (2004) notes that the introduction of the bow between 2100 and 2400 BP correlates with a rise in arrow-related injuries and defensive wounds. A growing concern with defense is reflected in profound changes to settlement and land use strategies, potentially leading to the emergent village nucleation of this period, characterized by larger house clusters, concealed food storage, and mesa-top fortified sites (Chatters, 2004).

Southeast

In the North American Southeast, bow and arrow projectile systems may have arrived as early as 3000 cal BP, as large, stemmed points declined and small, triangular points appeared (Daniel, 2021:152). The timing of this transition varies, though, within the Southeast and across the Eastern Woodlands generally, for which region several chronologies have been developed (Bettinger & Eerkens, 1999; Blitz, 1988; Bradbury, 1998; Nassaney & Pyle, 1999). Some of the earliest syntheses of southeastern archaeology (Caldwell, 1957; Coe, 1964) report that the shift from stemmed points (likely used with atlatls) to small triangular points coincides with other technological changes like the production of ceramics at the end of the Late Archaic period ca. 4000 cal BP (Odell, 1998). Still, while small triangular points are present in some Late Archaic records, the widespread decline of dart points and proliferation of small, triangular arrow points really characterizes the Late Woodland period, from 1250 cal BP until 950 cal BP (Nassaney & Pyle, 1999). “Dart” and “arrow” are sometimes used interchangeably in the southeast (Odell, 1998), though the difference between these technologies is noted by Lyman and colleagues (2008). Recent work by Hensen maintains a distinct difference between early triangular

forms (called "darts") and later triangular forms (asserted to be arrows) based on technological differences between Early and Late Woodland forms, ethnohistoric accounts, and rock art evidence (2025:49–53).

It is argued that, even with the introduction of the bow, atlatl technologies persisted in the Eastern Woodlands through the Late Woodland period (Bradbury, 1998; Nassaney, 1992). Some suggest that the use of the bow was driven by increasing sedentism and the adoption of horticulture (Nassaney, 1992; Story, 1990). Major changes in ceremonial, mortuary, ritual, and political practices co-occur with these shifts in subsistence and mobility (Anderson & Mainfort, 2002). Peres (2017) notes that increased bow use coincides with early maize agriculture, though the spread of bow and arrow technology outpaced that of farming (Peres, 2017).

Anderson and Mainfort (2002) identify the Late Woodland as a period of pronounced cultural change across the Southeast, with population growth inferred from the increased frequency of small communities and households within them. With sedentism and demographic expansion, some argue for increased pressure on food resources, resulting in increased levels of warfare (Blitz, 1988; Milner, 1999; Nassaney and Pyle, 1999). Later in the Mississippian period (beginning ~ 950 cal BP), further elaboration of agriculturally focused villages and resource stress is reflected in the building of palisades for defensive fortification in coincidence with the ubiquity of bow and arrow technology (Keeley *et al.*, 2007; Krus, 2016).

Bow and arrow technology features prominently in these observations of increased warfare from the Late Woodland into the Mississippian, yet the shift to sedentary lifeways and increased resource stress is not the only explanation for the spread of the bow into the Southeast. Jackson and Scott (2002) also mention the effectiveness of bow and arrow technology in the procurement of small game, which could have been important for supplementing agricultural diets. Fiedel (2014) presents a unique perspective on projectile point change across the Eastern Woodlands, in which Bond events (*i.e.*, cooling events throughout the Holocene that occurred every ~ 1500 years) coincide with major changes in lithic technological organization and stylistic expression. Fiedel's synthesis does not focus particularly on bow and arrow technology but does offer a plausible explanation for the major shifts shown in projectile point chronologies from the Southeast (sensu Coe, 1964; Daniel, 2021).

The Great Basin and California

The adoption of the bow and arrow in the Great Basin and California roughly spanned the millennium between 2000 and 1000 cal BP. In the Great Basin, this adoption is marked mainly by the appearance of Rose Spring corner notched and Eastgate expanding stem points, both often subsumed under the "Rosegate" type (Thomas, 1981). These relatively small, triangular points appear derived, perhaps via trial and error learning, from earlier and larger Elko corner notched dart points (Bettinger & Eerkens, 1999; Justice, 2002). Direct radiocarbon dating of hafting materials on Rosegate points (Smith, *et al.*, 2013) and organic weaponry (Rosencrance, *et al.*, 2023) places the earliest bow use in the Great Basin ca. 2000 cal BP, with widespread establishment by 1500 cal BP. Importantly, atlatl/dart

systems did not disappear immediately; dart points are found in strata dating to as late as 845–500 cal BP in places like Gatecliff Shelter and the Rose Spring site (Kennett, *et al.*, 2014; Thomas & Bierwirth, 1983; Yohe, 1998). Small, triangular, and side-notched Desert Series points largely replace Rosegate points in the Great Basin 950–600 cal BP, though Rosegate points continued to be manufactured during the first five or more centuries of this span (Smith *et al.*, 2013). It is thought by some that the spread of Desert Series technology marks the intrusion of Numic-speaking groups from southeastern California (Delacorte, 2008).

In California, the earliest arrow points are Rosegate points found east of the Sierra crest, within the hydrographic Great Basin; these date to 1750–1450 cal BP (Yohe, 1998). Arrow technology appears to have spread west from the Great Basin and into California proper, with Rosegate points found in sites on the western slope of the Sierra Nevada that date to as early as 1650–1350 cal BP (Kipps 1982; Moratto, 1972). The connection between Great Basin-derived arrow points and the more diverse types of California's Central Valley and coastal regions is uncertain, but it appears that the adoption of the bow along the coast occurred much later in time than it did in cismontane California. Leaf-shaped arrow points that were perhaps derived from earlier leaf-shaped atlatl darts do not appear in southern and central coast contexts until 1300–1050 and 1050–850 cal BP, respectively (Codding & Jones, 2007; Pletka, 2001). Barbed arrow points first appear 1050–900 cal BP in the northern Sierra Nevada foothills (Milliken *et al.*, 2007), and distinctive serrated arrow points appear in the Sacramento Delta and San Francisco Bay Area by 900–650 cal BP (Groza *et al.*, 2011). Combined, this information suggests the spread and adoption of bow and arrow technology in California was geographically variable and took nearly six centuries, from perhaps 1500 cal BP in the Sierra foothills to 900 cal BP or later near San Francisco. As in the Great Basin, dart systems in California appear to have persisted for many centuries while the bow was being adopted (Yohe, 1998), but contravening evidence from the Central Valley suggests the shift from dart to arrow was rapid, radical, and pronounced (Rosenthal, 2021).

Though Bettinger (2015) hypothesizes that the late adoption of the bow along California's coast may have been linked to the area's coastal and marine ecology, the shift from dart to arrow systems across the region does not appear to be intrinsically or causally linked to the environment (Kennett *et al.*, 2013). Environments across the region range from arid salt flats and steppes to alpine mountains, to estuaries and rocky coastlines, suggesting no one environmental characteristic favors one technology over the other. Similarly, this technological shift spanned both the relatively mesic and ameliorated climatic conditions following the Late Holocene Dry Period of 3000–1800 cal BP (Mensing *et al.*, 2023) as well as the multidecadal megadroughts interspersed with brief mesic periods that characterized the Medieval Climatic Anomaly of 1100–650 cal BP (Hughes & Diaz, 1994), suggesting climate was also not a primary driver of the bow's adoption.

Rather, the bow's adoption in the Great Basin and California appears to have been linked to changing settlement, economic, and social systems. In the Great Basin, shifts from more residentially stable Fremont farmers in the east, Lovelock foragers in the west, to more residentially mobile, family-band-oriented social systems

appear affiliated with the evolution of an intensified seed and nut-oriented subsistence economy (Bettinger, 1999).

A nearly identical economic shift occurred in California except that, due to demographic packing, residential mobility was much more constrained. Economic decision-making, like in the Great Basin, nonetheless also shifted for the most part to individual households within California's generally stratified and complex but also anarchic, tribelet-based sociopolitical organization (Bean & King, 1974; Kennett *et al.*, 2013; Kroeber, 1925). It has been argued that the bow was essential to this transition, by allowing individually (as opposed to communally) oriented hunters to share the meat required among most hunter-gatherer societies. This allowed meat-sharing households to privatize the collection and storage of costly-to-process seed resources, the intensive labor for which precluded large-scale sharing (Bettinger, 2015).

Left unexplained in this scenario is not why the bow was adopted, but why it was not, especially in terms of why such an effective technology took so long to catch on nearer the coast. Perhaps environments that were less conducive to hunting had something to do with it, but the fact that pre-bow groups in and around the San Francisco Bay Area lived in large, centralized societies marked by very large shell mounds arguably suggests that the large, prestige-oriented, redistributive economies that likely dominated pre-bow proved particularly resistant to the privatization brought about by the adoption of such a radically disruptive technological system (Hildebrandt & McGuire, 2002; Lightfoot & Luby, 2002).

Southwest

The timing of the bow and arrow's arrival in the US Southwest differs across the region but is generally believed to have occurred between 1750 and 1200 cal BP (Reed, 2010; VanPool & O'Brien, 2013). Some of the earliest evidence comes from Glen Canyon (southeastern Utah) and southwestern Colorado (*e.g.*, the Tamarron and 5DL896 sites), where small point forms suggest bow use as early as 1750–1650 cal BP (Reed, 1990). By Basketmaker III (ca. 1450–1200 cal BP), the bow was widespread, especially in the northern Puebloan Southwest, and gradually diffused southward into Mogollon territory over subsequent centuries (VanPool & O'Brien, 2013). Across the region, it was the dominant projectile system by about 1250 cal BP (Roth *et al.*, 2011), although mixed assemblages of dart and arrow points from the bow's introduction through the Basketmaker III period (to ca. 1200 cal BP) indicate a protracted period of technological overlap.

The uncommon preservation of organic material in the arid Southwest permits a more detailed assessment of bow and arrow construction than is possible in other regions. The technology was comparatively simple in the Southwest, consisting of relatively straight, short bows made from a single piece of wood (self-bows) until roughly 750 cal BP, when these were replaced by sinew-backed, recurved bows (LeBlanc, 1999; VanPool & O'Brien, 2013). Arrows were predominantly untipped, single pieces of wood (self-arrows) or a composite of a wooden foreshaft and a cane or reed shaft. The striking end was typically blunt or sharpened to a point (Hamm,

1991; Haury, 1950); only occasionally were arrows tipped with stone and perhaps particularly in the context of warfare, where the deeper penetration afforded by stone tips may have been desirable (VanPool, 2003; Waguespack *et al.*, 2009).

Granting that hunting returns may have improved with the adoption of the bow and arrow, Southwestern economies were already centered on maize agriculture by that time, which has led many scholars to focus primarily on the technology's social role. Widespread use of the bow in the Southwest correlates with regional population growth, especially after 1350 cal BP when evidence suggests increasing numbers of larger, denser settlements and more complex social organization.

VanPool and O'Brien (2013) evaluate how the adoption of the bow influenced sociopolitical complexity in the ancient American Southwest, assessing the relationship between weapon technology, warfare, and forms of social organization across the region. They conclude that the relationship between bow adoption and social complexity is real but variable and context-dependent: bow-subsidized warfare drove sociopolitical complexity in some areas while, in others, the technology lowered the cost of punishing defectors, thereby supporting the formation of larger, more complex societies. In some contexts, it contributed to the growth of sociopolitical complexity through coercion and enforcement (as in Chaco), but they conclude, it was neither necessary nor sufficient for complexity to arise.

Continental Synthesis

Granting the general north–south temporal trend evident in Fig. 1, the trajectory of bow adoption across North America was neither uniform nor linear. Regional variation suggests that the transition was shaped not only by environmental and functional considerations, but also by deep structural differences in social organization and cultural transmission (*e.g.*, Shea, 2017). We identify three cross-cutting axes of variation that together help to define the contours of adoption.

Coexistence with Other Projectile Systems

The bow rarely replaced the atlatl immediately. Indeed, the two systems coexisted for centuries—sometimes millennia—across much of the continent; the bow's superior performance in many regards did not automatically render older systems obsolete. In coastal Arctic settings, where the bow complemented harpoons and atlatls and never fully replaced them, the overlap appears to have been ecologically justified. This may also be true of other regions, but in the Southeast, for example, protracted co-occurrence suggests that, beyond ecological or functional considerations, the transition required changes in social norms, knowledge systems, and identities. That is, across the continent, adoption was shaped not only by economics, but also by the fit between weapon performance characteristics, local prey structure, group mobility, and the social meaning of weaponry itself.

Social Context of Adoption

While ecological and demographic factors influenced the bow's utility, its uptake was consistently tied to shifts in social organization. In every region we discuss, the bow's spread aligns with shifts in group size, levels of conflict, and/or increased need for extra-familial cooperation—often more so than with potential ecological triggers, as has been argued by many of the authors on whose work we draw. In the Southwest, for example, the bow facilitated the expansion of large, sedentary villages, perhaps by lowering the cost of enforcing cooperation within growing populations (Bingham & Souza, 2009). In several regions (*e.g.*, Arctic, Pacific Northwest, Southeast), its adoption coincided with intensified conflict, territoriality, and the emergence of fortified settlements. These correlations suggest that the bow was sometimes as much a sociopolitical instrument as it was a hunting implement.

Ecological Plasticity vs. Social Selectivity

The bow's ecological versatility is one of its most distinctive features: it was adopted in environments ranging from Arctic tundra to coastal estuaries to desert basins. Yet the timing, rate, and mode of adoption were highly uneven. It was adopted early in the Arctic despite material constraints on both bow and arrow construction, and late on the California coast, where the bow's support of individual autonomy flew in the face of centralized redistribution systems and prestige economies. It spread rapidly among mobile economies of the Great Basin and parts of the Plateau, where it aligned with intensified, seed-based subsistence and smaller social units. These differences show that adoption was less about physical environments and more about social architectures. In the next section, we explore in more detail the idea that social selectivity was the primary filter through which ecological opportunities were interpreted: even when a tool would be useful in a given environment, social norms, political organization, and economic institutions determine whether it is accepted, resisted, delayed, or modified.

A Model of Cultural Transmission and Technological Intensification

Documenting the geographical spread of technology is a fundamentally empirical exercise, but for reasons described above, interpreting documented patterns requires modeling both the ecological and social contexts of adoption. To this point, we have primarily considered the benefits of bow adoption. Importantly, though, the bow was also costlier to produce than existing weapons systems, making its adoption an instance of technological intensification, which we define as a shift toward tools or practices that require greater upfront investment (*e.g.*, time, expertise) but yield increased efficiency, control, or returns per unit of effort. A previous model of technological intensification applied to this problem (Bettinger *et al.*, 2006) is

not only flawed but also does not adequately account for the agents of transmission. It matters, for instance, whether technological intensification is driven primarily by craft specialists or do-it-yourself tool maker-users, a difference likely germane to the spread of the bow and arrow.

Craft Specialists

The logic of tool choice for the craft specialist is captured by revising Bettinger and colleagues' (2006) model of technological intensification, which wrongly argues that the switch from an expedient (*i.e.*, quicker to make, less effective) technology to a more effective but labor-intensive technology—say from atlatl to bow—will occur once the time spent using those technologies exceeds some common critical use time. Unfortunately, when structured this way, the model makes manufacturing time a benefit-producing cost, which is correct (manufacturing time increases return rate), but uses time as a non-benefit-producing cost, *i.e.*, a cost pure and simple, which is not correct: use time obviously generates benefits. The original model makes perfect sense, however, if cast from the perspective of a craft specialist for whom the common switching time threshold is overhead, *i.e.*, fixed or base costs of tool production, whether the craft specialist is making either the expedient or more costly technology. Overhead might include the cost of acquiring and maintaining the tools, knowledge, and skills needed to engage in tool production; of the facilities where tools are produced or stored until distributed; and of establishing credibility as a specialist or acquiring necessary permissions to produce tools for others. Craft specialists will aim to optimize the exchange value of their products in relation to overhead and manufacturing time, where the return rate obtainable using the product is a reasonable proxy for exchange value:

$$\frac{r_i}{m_i + o}$$

where

r_i is the return rate in kcal/h obtainable with a given technology, a proxy of its exchange value.

m_i is manufacturing time—time (hr) spent making a particular kind of procurement or processing technology i.

o is production-related overhead—fixed or base costs of tool production, whether one is making either the expedient or more costly technology.

On this view, a lower return, expedient technology is viable for craft specialists so long as its manufacturing efficiency—the ratio of its return rate to manufacturing time r/m—is greater than that of a more costly, higher return alternative, and there are no overhead costs. As overhead costs grow, however, craft specialists will increasingly shift production to the more costly, higher return technology, for

which they can expect to obtain correspondingly more in return. The overhead time switching point, at less than which the more expedient, lower return technology (m_1) is favored, and above which the more costly, higher return technology (m_2) is favored, is as follows:

$$o_{1\leftrightarrow 2} = \frac{r_2 m_1 - r_1 m_2}{r_2 - r_1}$$

Do-It-Yourself Tool Producer–Users

Whereas craft specialists rationalize production to optimize return rates (exchange value), do-it-yourself tool producer–users (DIYers) rationalize production to optimize actual returns (return rate multiplied by use time), in this case relative to overhead, manufacturing, and use time:

$$\frac{r_i u_i}{m_i + o + u_i}$$

where

o, m, and r	are overhead, manufacturing time, and return rates, respectively, as defined above.
u	is use time—time expended using the technology for procurement or processing expressed in hours (hr).

Because DIY tool producer–users are returns oriented, it is useful to define the point at which they should switch from a cheaper, lower return technology to a higher return but more costly technology in terms of total expected return, at less than which the cheaper technology, and more than which the more costly technology, is favored:

$$v_{1\leftrightarrow 2} = \frac{m_2 - m_1}{{r_1}^{-1} - {r_2}^{-1}}$$

where

r and m	are defined as above.
v	is the total return in kcals, kg, or li.
$v_{1\leftrightarrow 2}$	is the total return switching point, at less than which the less costly to manufacture technology (m_1) is favored, above which the more costly (m_1) is favored.

While neither use nor overhead times figure directly into this total return switching point, use time is obviously implied and easily obtained by dividing the total return value by return rate, obviously differing by technology. Overhead time cannot

be obtained this way because it has no effect on the DIYer's total return switching point: lay toolmakers are indifferent to differences in overhead time.

Model Implications and Applications in North America

Our model clarifies why the bow and arrow may have diffused unevenly—sometimes rapidly, sometimes slowly—depending on the social and economic roles of tool producers and users. Importantly, it also shows that the technological choices of craft specialists and DIYers are at odds with some intuitions regarding their behavior. Regarding expedient technologies, for instance, craft specialists will never produce a lower return technology whose manufacturing efficiency, r/m, is inferior to that of a more costly but higher return alternative; DIY producer–users will resort to any technology that minimizes time to reach a target return regardless of manufacturing efficiency. Secondly, increasing overhead favors shifts to more costly, higher performing technologies by craft specialists (Fig. 3) but not tool users, for whom it does not affect total return switching point (Fig. 4). Neither result has anything to do with inherent differences in skill (*i.e.*, m) but is simply a function of cost-benefit structures that govern the rewards sought by craft specialists and lay producers.

California Delta and San Francisco Bay Regions

The tardy appearance of bow and arrow technology in the California Delta and San Francisco Bay regions (ca. 900 cal BP) is likely a reflection of its initial spread in primitive (*i.e.*, initial) self-bow form among do-it-yourselfers for whom its adoption hinged on expected levels of return that were too low in those regions to promote the shift. Its later Delta–Bay area adoption is probably not a reflection of change in the overall demand/supply of hunted products, but rather, increasing Late Period

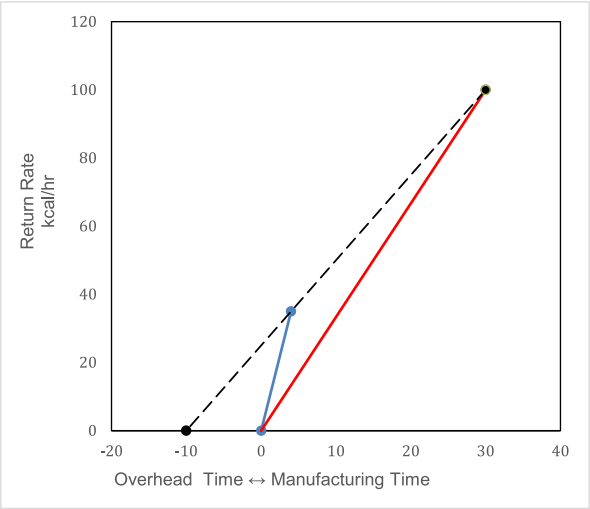

Fig. 3 Given a cheaper to make ($m = 4$ h), lower return ($r = 35$ kcal/h) technology (blue line) and more costly to make ($m = 30$ h), higher return ($r = 100$ kcal/h) technology (red line), the craft specialist will switch from the less to more costly at when overhead costs exceed 10 h

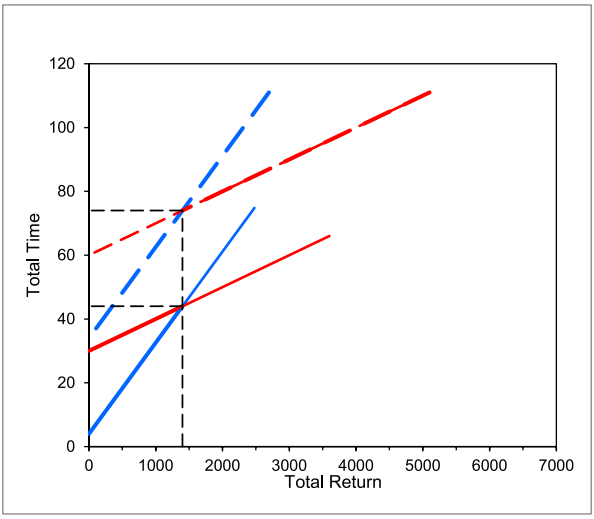

Fig. 4 Given a cheaper to make ($m = 4$ h), lower return ($r = 35$ kcal/h) expedient technology (blue line), and a more costly to make ($m = 30$ h), higher return ($r = 100$ kcal/h) technology (red line), the do it yourselfer will switch from the less to more costly technology when the required total return exceeds 1400 kcals whether there is no cost in overhead time (solid lines) or 30 h of overhead time (dotted lines)

craft specialization. Hunter-gatherers across California manufactured products like moccasins, baskets, snowshoes, arrowheads, and shell beads to participate in well-developed local and regional trade networks (Arnold, 1987, 1992; Bennyhoff & Hughes, 1987; Davis, 1961; Eerkens & Spurling, 2008), and bow-making specialists are well documented in the region (*e.g.*, Davis, 1961; Gifford, 1926; Goldschmidt, 1951; Powers, 1877). Craft specialization in the Delta–Bay area was associated with the development and use of money, which supplanted barter for the exchange of an increasing variety of craft products (Bettinger, 2015). So, as noted above, as hunter numbers decreased and bow/arrow maker numbers increased, the expected/required returns of the former and overhead costs of the latter would both have increased, favoring a shift to the most costly but effective bow technology.

Southwest

Adoption of bow and arrow technology—in the form of simple self-bows and self-arrows—was also relatively late in the Southwest (ca. 1750 cal BP), and uptake was protracted. At the time of first introduction, craft specialists of various sorts were likely present across the Southwest (*e.g.*, Brown and Paddock, 2011; Crown, 2014), but subsistence economies centered on maize agriculture and communal (atlatl) hunting of large ungulates. In this context, the bow might have been adopted initially by (DIY) individuals on the margins of cooperative labor (*e.g.*, young men, peripheral households) looking to increase returns on individual/household-level hunting of small game. Its widespread use lagged for ~ 400 years and eventually took root in the context of population growth and social reorganization. During Basketmaker III and Pueblo I, the conditions of social aggregation and intensified agriculture may have increased the need for (cost of) maintaining order and ensuring cooperation among non-kin, and communal labor (farming and other infrastructural construction and maintenance) created shared obligations. This created a scenario in which the

bow provided a return on investment, *v*, not in calories but in social stability. While significant, these returns may have remained too low to warrant investment in costlier-to-produce sinew-backed, recurved bows until ca. 750 cal BP (Pueblo III), when social tensions may have peaked, exacerbated by prolonged drought.

Model Versatility

The measure of a model's worth is not its universality but its ability to clarify variation and reveal the conditions under which different outcomes emerge. So, while we do not mean to suggest that our transmission model will explain each instance of bow adoption, we do count broad applicability and versatility among its strengths.

As in the Southwestern case, variables can be redefined according to local contextual evidence. It is possible, for instance, to connect the craft specialist and DIYer models by expanding the definition of manufacturing time, *m*, for the DIYer to mean time spent making a particular technology or amassing material equivalent to its value in trade, and the definition of overhead, *o*, for the craft specialist to mean indirect costs broadly, including not only the base costs of tool production but also time diverted to craft production from everyday activities essential to individual support (*e.g.*, hunting, food preparation, trade, *etc.*). This connection allows technological intensification via specialization without any change in basic demand. That is, the demand for more costly but better-performing tools will increase as (1) tool users increase production to obtain those tools in trade rather than by making them, and (2) the craft specialists divert time to craft production they formerly spent supporting themselves on other fronts.

Discussion and Conclusions

Adoption of the bow and arrow in North America had far-reaching implications for economic behaviors, interpersonal negotiations (conflict, cooperation), and social organization. While environmental variables such as wood availability and prey diversity/behavior were surely important, demographic and cultural variables—including population size, density, and connectivity; cultural transmission processes; and existing social dynamics—were at least equally influential. The different decision parameters for craft specialists versus DIY producers–users articulated in the modeling section above provide a useful example, explaining both early and delayed adoption across regions. Regions with specialist economies (*e.g.*, stratified, redistributive societies) may have delayed adoption due to resistance to decentralizing technologies. In mobile or household-based economies, the bow may have been adopted more quickly if it aligned with individual autonomy and flexible subsistence strategies. The model has clear potential for application beyond North America. With careful attention to archaeological indicators of craft specialization, group size, mobility, and conflict, our model offers a powerful explanatory framework for the diverse tempos and trajectories of bow adoption across the world.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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